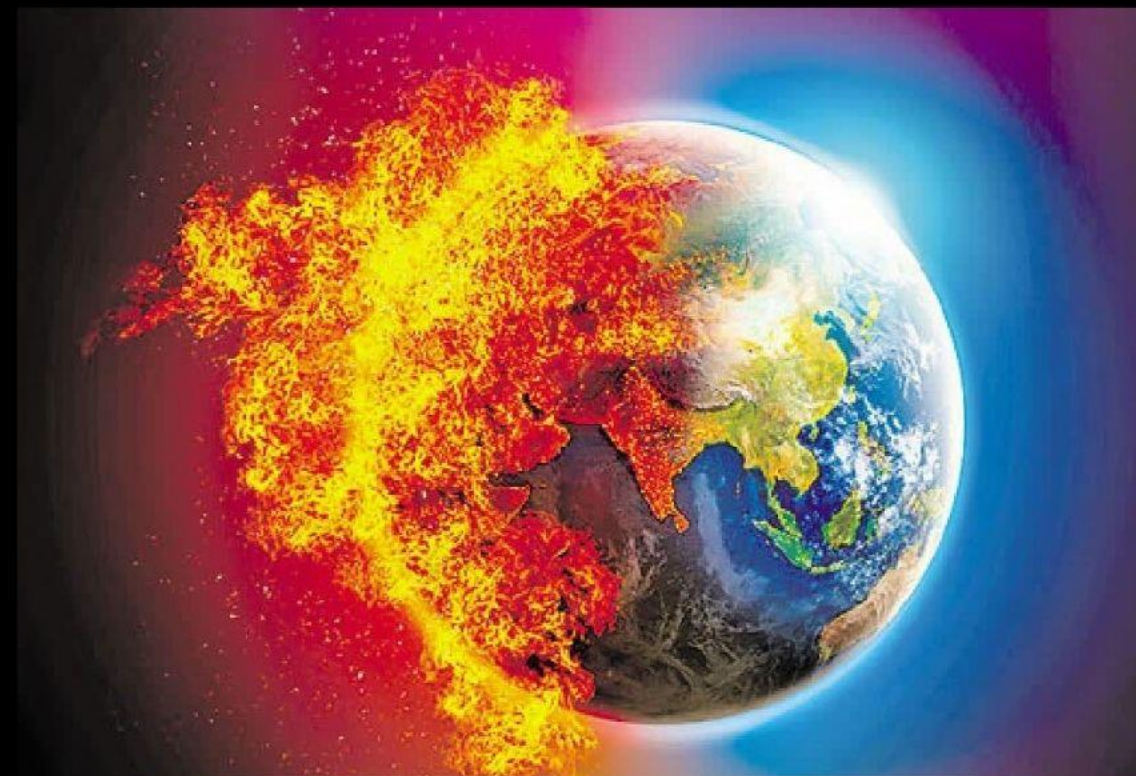


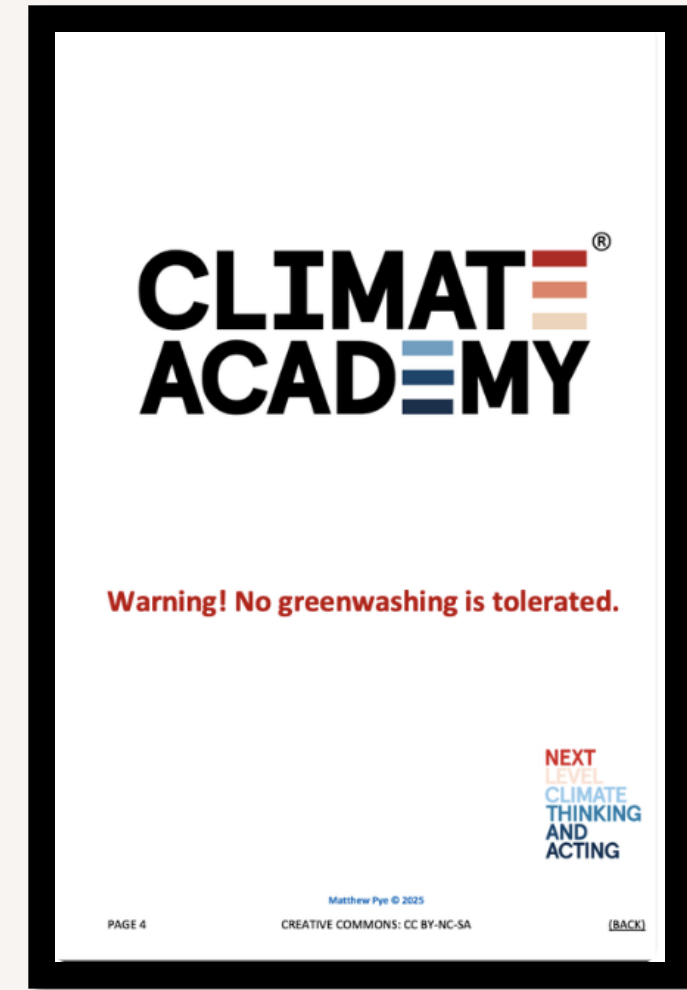


From Text to Knowledge Graphs: SemanticClimate



Made by: Amyra Jasuja, Volunteer at
SemanticClimate

-  Climate Academy is a student book that connects climate, science, philosophy, economics and politics to one another.
-  It aims at creating a deep and accurate understanding of climate change so that the intensity of the problem at hand is understood and the right action is taken.
-  Furthermore, the book focuses on empowering students to face climate anxiety with courage and resilience in order to overcome the problem at hand.



OUR GOAL:



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Resource depletion and environmental shift is taking place at an unprecedented rate.



Students need to understand the impact and effect of climate change.



At Semantic Climate, we aim to make this knowledge not only accessible but fully comprehensible.



Thus, every student can make the environment a priority and participate actively in alleviation of this climate emergency.



Desertification in Haryana, India.

SO, HOW DID WE DO IT?

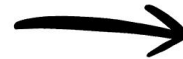


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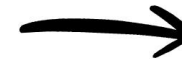
Workflow At SemanticClimate:



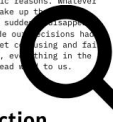
Source Text: Climate Academy



Converting the PDF into a TXT file



Such moments do not have to be religious of course. Probably the most common experience of finding synergies in the world happen when we find ourselves in a loving relationship. These relationships often don't make much sense from the outside. Why him, why her? There is rarely a rational explanation for why we fall in love. Sometimes love is found with people who are utterly different to us. There are other places that we can find synergy of course: the starry night, a deep sunset, a moment of laughter in a group where you suddenly notice that you belong, a book, or even sometimes just the wandering mind on a bus that settles into a place of understanding. These symbolic moments can also happen reverse. A loss of meaning can be abrupt, usually for tragic reasons. Whatever the cause, the bright stars that make up the constellations of our life, can fade or suddenly disappear. The map of meaning that we used to guide our decisions has made so much natural sense, but it can get confusing and false. Like in that clearing by the river, everything in the world can revert to looking like dead wood to us.



Keyword Extraction



Creation of the Climate Academy Encyclopaedia



1	keyword	
2	climate change	
3	climate crisis	68
4	Plato	61
5	Climate Academy	55
6	students	54
7	capitalism	47
8	India	45
9	democracy	38
10	Academy	29
11	media	29
12	Rousseau	29
13	history	28
14	CUT's Index	26
15	Anthropocene	26

Structured Keyword List (CSV)



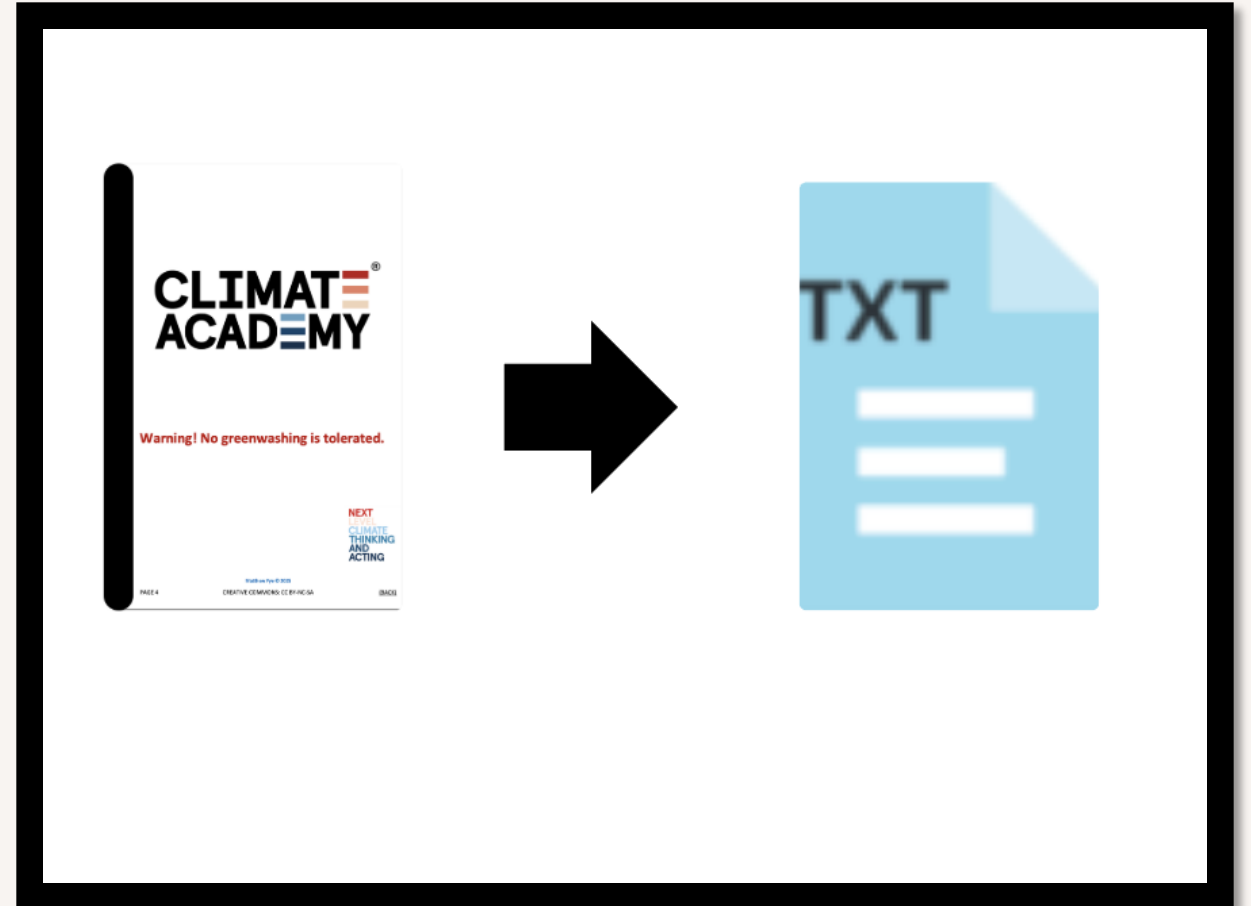
STEP 1: CONVERTING IT INTO A TXT FILE:



To create our encyclopaedia and knowledge graph, we first convert the given text into a TXT file.



Here, the PDF of the Climate Academy student book was taken and converted into a TXT file in order to extract the keywords to ultimately build our dictionary using various tools on Python.



STEP 2: EXTRACTING KEYWORDS:



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- At SemanticClimate, we use Python to extract keywords from Climate Academy which are initially arranged in decreasing order of frequency.
- It is from there that the terms are arranged alphabetically in order to create the encyclopaedia. This helps organise the text further and create easy to understand knowledge.

1	keyword	count
2	climate change	97
3	climate crisis	68
4	Plato	61
5	Climate Academy	56
6	students	54
7	capitalism	47
8	India	45
9	democracy	38
10	Academy	29
11	media	29
12	Rousseau	29
13	history	28
14	CUTx Index	28
15	Anthropocene	26

List of the top 15 most reoccurring terms in "Climate Academy" given in decreasing order of frequency.

STEP 3: CREATION OF THE ENCYCLOPAEDIA:



After further structuring the extracted keywords into a list, we can successfully build a dictionary.



This enables us to create the encyclopaedia, thus making the text concise and understandable for all.

climate change

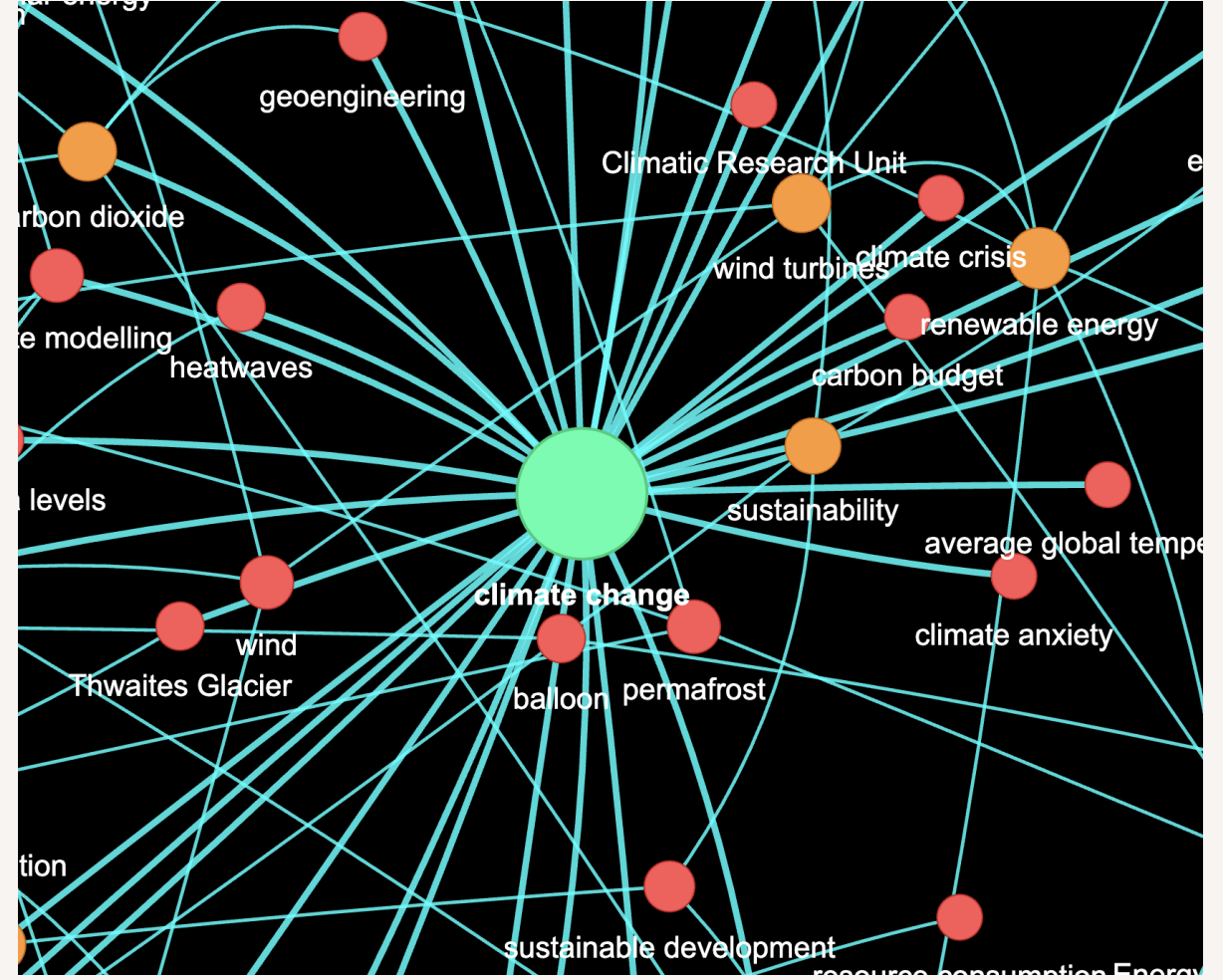
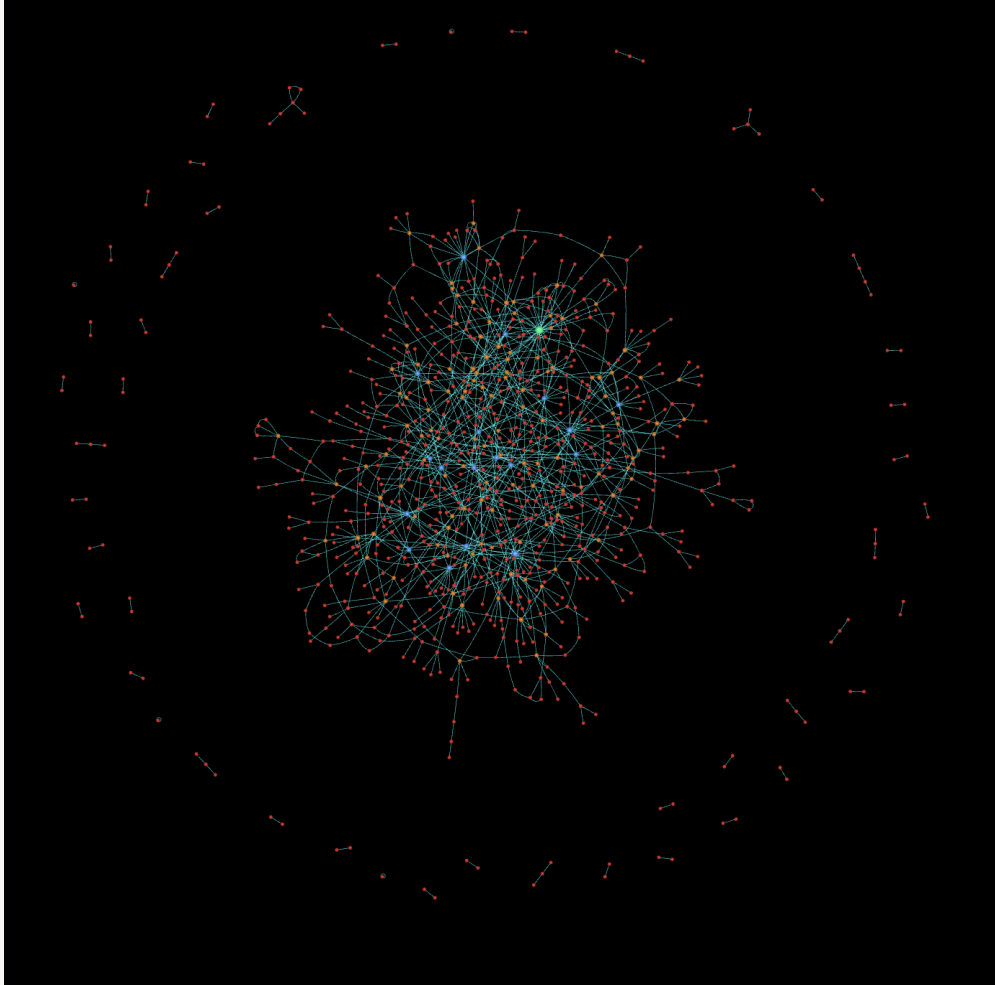
search term: climate change [Wikipedia Page](#)

Present-day **climate change** includes both **global warming** —the ongoing increase in global average temperature —and its wider effects on Earth's climate system . Climate change in a broader sense also includes previous long-term changes to Earth's climate . The current rise in global temperatures is driven by human activities , especially fossil fuel (coal , oil and natural gas) burning since the Industrial Revolution . Fossil fuel use, deforestation , and some agricultural and industrial practices release greenhouse gases . These gases absorb some of the heat that the Earth radiates after it warms from sunlight, warming the lower atmosphere. Carbon dioxide , the primary gas driving global warming, has increased in concentration by about 50% since the pre-industrial era to levels not seen for millions of years.

An example of one of the terms given in the Climate Academy Encyclopaedia.



FINAL STEP: CREATION OF THE KNOWLEDGE GRAPH:





1. Kyrene Solanki, 11th Grade Student at Vasant Valley School: " I found the presentation very insightful. I particularly liked the Knowledge Graph as it helped me establish connections with climate change that I was unaware of. This visually highlights the impact of climate change and makes it very relatable."

2. Reet Bhatia, 12th Grade Student at The Shri Ram School, Aravali: " The presentation was very exciting. I learnt how many of my interests are interlinked with climate change and that makes me want to take action."

3. Anaisha Moudgil, 11th Grade Student and Secretary of Environment at The Shri Ram School, Aravali: "I really liked the concept of bringing awareness on such a large scale. By relating various concepts of deep interest, many will now feel a sense of motivation to understand the problem and find a solution to save the world."

REFERENCES AND SOURCES:



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Climate Academy Encyclopaedia:

[https://vivliostyle.org/viewer/#src=https://github.com/semanticClimate/demo_CAbookEncyclopedia/blob/main/manifest.jsonld&f=epubcfi\(/2!/4/10/28\[Adaptive Cycle\]/8\)](https://vivliostyle.org/viewer/#src=https://github.com/semanticClimate/demo_CAbookEncyclopedia/blob/main/manifest.jsonld&f=epubcfi(/2!/4/10/28[Adaptive Cycle]/8))

Climate Academy Student Book:

https://github.com/semanticClimate/demo_CAbookEncyclopedia/blob/main/ClimateAcademy_Book.pdf

Climate Academy Knowledge Graph:

https://github.com/semanticClimate/demo_CAbookEncyclopedia/blob/main/CA_encyclopedia_KG.html

Climate Academy Keyword List:

https://github.com/semanticClimate/encyclopedia/blob/renu/CAbook_encyclopedia/All3088_CAbook_words.txt